

Google.org Impact Challenge: AI for Science

I. Organization and Submitter Info

1. What is the full, registered (legal) name of your organization?

Yale University

2. In which country is your organization headquartered?

United States (including: PR, GU, VI, AS, MP)

3. Which of the following classifications applies to your organization?

University/Research Institution: public or private academic or research institution

4. Please share the following information about your organization.

For a research lab within an academic institutions, please use the information for your research lab, not the academic institution to complete the following questions.

4.a. What year was your organization founded? Enter as YYYY.

2014

4.b. What is your organization's current annual budget (approximate in USD)?

\$785,000

4.c. What is your organization's current number of full time employees?

12

5. If your organization has a website and/or a primary social media channel, please provide the links below.

5.a. Website

<https://www.comitalab.com/people.html> (<https://www.comitalab.com/people.html>)

5.b. Primary Social Media Channel

n/a

6. Have you received funding from Google.org before?

No

7. How did you first discover or hear about this funding program and opportunity?

- Website Search

8. Primary Point of Contact

Please identify the individual who will serve as the primary point-of-contact for your organization throughout the Google.org (<http://Google.org>) Impact Challenge: AI for Science.

8.a. Full Name

Nohemi Huanca-Nunez

8.b. Job Title

Associate Research Scientist

8.c. Email

nohemi.huanca@yale.edu (mailto:nohemi.huanca@yale.edu)

8.d. Timezone

UTC -5, EST, New York, Toronto, Lima, USA/Canada (Eastern), Peru

9. Secondary Point of Contact

Please provide a secondary contact person who can be reached if the primary lead is unavailable.

9.a. Full Name

Liza Comita

9.b. Job Title

Davis-Denkman Professor of Tropical Forest Ecology

9.c. Email

liza.comita@yale.edu (mailto:liza.comita@yale.edu)

9.d. Timezone

UTC -5, EST, New York, Toronto, Lima, USA/Canada (Eastern), Peru

10. Please note that all interviews, follow-up communications, and post-funding support from the Google.org Impact Challenge: AI for Science will be conducted in English. Can you confirm that at least one project lead from your organization is fluent in English?

Yes

II. Impact

11. What is the title of your proposal?

SeedLearn: Teaching AI to Think Like a Botanist for Plant Species Identification

12. Please select the topic(s) that best align with your project's scope.

- Climate Resilience & Environmental Science - Biosphere, Climate, and Society
- Climate Resilience & Environmental Science - Food & Agriculture

13. Please select the open-source output(s) that best align with your project's scope and goals.

- Peer-reviewed publication(s) (Open Access preferred)
- Curated Dataset(s) (CC0 or CC-BY 4.0)
- Software, Tools, or Platforms (Apache 2.0 or MIT license)
- AI Models or Architectures (OpenRAIL or Apache 2.0)

14. Please select the geographic regions your project serves.

- Global

15. Please specify your project's current stage.

Proof of Concept / Preliminary Results: Initial experiments, models, or pilot studies have been conducted and demonstrate feasibility.

Please complete the following sentence prompts.

16. Problem Statement

16.a. The specific challenge this project addresses is...

Biodiversity loss is accelerating, increasing the need for accurate and scalable species identification. Current approaches rely on large, well-labeled image datasets (e.g., iNaturalist) and are inherently limited in real-world ecosystems where most species are rare and underrepresented. In diverse ecosystems (e.g., tropical forests), where hundreds of visually similar species co-occur, collecting sufficient labeled images is logistically infeasible, making image-only approaches unreliable. Yet, vast written botanical knowledge in field guides and taxonomic keys remains largely untapped.

16.b. This challenge is significant because...

This limitation creates a major bottleneck for biodiversity science, conservation, and restoration. Accurate species identification is foundational for monitoring ecosystem change, guiding management decisions, and assessing climate impacts, yet current approaches cannot scale to real-world diversity. Reliance on scarce taxonomic expertise further constrains progress, especially in understudied regions. Without scalable identification tools, large volumes of ecological data remain unusable, limiting our ability to track biodiversity loss and implement effective conservation strategies in rapidly changing environments.

16.c. The core research questions this project seeks to answer to advance/inform the field are...

Building on our prototype, we investigate how AI systems can integrate visual data with structured written descriptions to enable species identification that reflects expert reasoning and establishes a new paradigm for imaging-limited biological classification. We ask which traits can be reliably extracted from images and linked to text, how trait-based reasoning improves accuracy, interpretability, and generalization, and how human-AI interaction, including in situ observations (e.g., leaf odor), improves performance when key traits are not visible.

17. Proposed Solution**17.a. The solution we are proposing is...**

We propose a biologically grounded AI-system that performs species identification by reasoning over evidence such as morphological traits, rather than relying solely on image classification. SeedLearn extracts diagnostic traits from images, aligns them with structured botanical knowledge, and integrates evidence to generate interpretable predictions with uncertainty. By mimicking expert botanical reasoning, SeedLearn enables accurate identification under data scarcity and supports use by non-experts, enabling scalable biodiversity monitoring while establishing a generalizable paradigm for image-limited classification.

17.b. The tools, methods, or techniques we are planning to use are...

The system integrates three coordinated components: (a) vision-LLMs to extract morphological traits from multi-angle field images; (b) a fine-tuned CLIP model for few-shot classification of labeled species; and (c) a retrieval-augmented generation architecture searches a curated corpus of 5,000+ botanical trait descriptions. These components are combined within a reasoning framework that synthesizes evidence to produce interpretable predictions with uncertainty, trait-level explanations, and supports interactive input when key diagnostic traits are not observable.

17.c. The evidence we have that our solution, once implemented, would solve this problem is...

We have developed a proof-of-concept system using over 10,000 expert-curated field images spanning 168 plant species, representing a highly challenging, real-world dataset. Few-shot classification with BioCLIP2 provides a baseline but remains limited for visually similar species. Our preliminary results show that diagnostic traits can be extracted and used to support literature-based matching and LLM reasoning, improving candidate ranking and interpretability. These results demonstrate the feasibility of scalable, trait-based reasoning for species identification under data scarcity.

17.d. Beyond addressing the immediate problem, our use of AI will catalyze lasting impact through new scientific inquiry and/or establish a new field benchmark by...

Beyond plant species identification, this project establishes a new paradigm for AI in image-limited biological systems by integrating visual data with structured ecological reasoning. Each prediction generates standardized, image-linked trait data, transforming identification into a scalable data-generation process. This creates a continuously expanding phenotypic dataset that centralizes botanical knowledge, improves performance on rare species, and enables new research (e.g., ecosystem responses). It also enables non-experts to identify and report biodiversity, expanding participation in ecological monitoring.

18. End Beneficiaries

18.a. The specific populations or communities that will ultimately benefit from this work are...

The end beneficiaries include conservation practitioners, ecologists, restoration organizations, and applied land managers (e.g., farmers), as well as local field technicians, students, and community stakeholders. By improving species identification in image data-limited systems, this project addresses a key bottleneck in biodiversity monitoring, reduces reliance on scarce taxonomic expertise, and supports more informed decision-making (e.g., early detection of invasive plant species). This makes identification more accessible, including across research, conservation, restoration, and environmental management globally.

18.b. Our strategy for centering beneficiary voices and incorporating their feedback is...

Centering beneficiary voices will involve ongoing collaboration and co-design with field-practitioners, land managers, ecologists, and botanists engaged through pilot testing and real-world use. Users will test SeedLearn in field settings and provide structured feedback on accuracy, usability, and relevance. Engagement with students and local collaborators, including use in botanical field courses, will support iterative refinement. This continuous feedback loop will guide model improvement and ensure the system meets practical needs in biodiversity monitoring and restoration.

18.c. The potential reach is...

We anticipate direct engagement with 50–80 researchers, field practitioners, and students via pilot deployments in Latin America and North America, alongside partnerships with 3–5 conservation and restoration organizations in the first 24 months. Direct impact will expand through workshops and co-development with practitioners and land managers. Broader reach through open-source platforms (e.g., GitHub, Hugging Face), conferences, and publications will extend to 500–2,000 users globally across ecosystems.

19. Expected Outcomes

19.a. The specific metrics we will use to demonstrate our solution's real-world impact are...

We will assess impact through improvements in identification accuracy (top-1 and top-5) relative to image-only baselines, particularly in image-limited, highly diverse, real-world field conditions. We will measure reductions in identification time and improvements in user performance, including the ability of non-experts to accurately use the system in practice. Impact will also be assessed through adoption across research, practitioner, and conservation communities, frequency of system use, and engagement with open science platforms (e.g., GitHub, Hugging Face).

19.b. The specific signals or negative indicators that would demonstrate our solution is failing to achieve its intended impact are...

Failure would be indicated by limited improvement in identification accuracy or no meaningful reduction in identification time in real-world-use. Low adoption across research, practitioner, and conservation communities, or lack of integration into existing workflows, would signal weak impact. Additional indicators include low user trust, poor performance in highly diverse field conditions, and inability of non-experts to use the system effectively. These signals would indicate the solution is not improving accessibility or supporting decision-making in practice.

19.c. The changes from baseline we expect to see are...

Current image-based models perform poorly in high-diversity, data-limited systems (often <1% top-1 accuracy), highlighting the difficulty of the task. We expect to substantially improve identification performance through multimodal, trait-based reasoning, targeting >70% top-1 and >85% top-5 accuracy. In parallel, we will scale from hundreds to thousands of individuals and expand coverage to hundreds of species, while enabling reliable trait extraction and interpretable, decision-support outputs for real-world biodiversity monitoring.

III. Innovative Use of Technology

20. What existing technologies will you use to build your solution, and what role will they play?

We will use vision-language models (e.g., Gemini) to extract morphological traits from multi-view images and support reasoning over structured evidence. A fine-tuned CLIP model will be used for candidate species ranking, with Gemini Embeddings 2 improving few-shot identification through enhanced similarity representations. A retrieval-augmented generation (RAG) framework, using embeddings and search (Vertex AI), will access curated botanical knowledge from field guides and trait databases. These components will be integrated within a scalable cloud-based system (e.g., Gemini APIs, Google Cloud Storage, GPU infrastructure) to enable multimodal, trait-based reasoning under image-limited conditions. We will iteratively update components as improved models become available.

21. If applicable, why is a new, custom-developed solution necessary?

Existing plant identification systems rely on large-scale image classification, which does not scale to hyperdiverse ecosystems (e.g., tropical forests). These approaches require extensive labeled data per species, yet most species are rare and poorly represented. In communities with hundreds of visually similar species per hectare, it is logistically impossible to obtain sufficient images for all taxa. As a result, image-only approaches are fundamentally constrained and will systematically fail in high-diversity systems. Our SeedLearn approach addresses this limitation by integrating visual data with structured textual biological knowledge and trait-based reasoning, enabling identification under extreme data scarcity and improving generalization across contexts.

22. Do you have an existing dataset being used for your solution?

Yes

If YES, please describe the critical datasets you will use for training/validation.

We have an existing curated dataset of over 10,000 plant images representing ~160 species, with multiple images per individual capturing different views. Because this dataset is embedded within long-term ecological research programs, species identities are expert-validated, and metadata include specimen identifiers and site information. The dataset currently spans multiple field sites in Panama and is expanding to additional regions (e.g., Ecuador) to improve generalizability. We are also compiling structured trait data from botanical keys and floras. Through established research collaborations, we have access to additional expert-validated data for thousands of individuals across species and environments, enabling continued model development.

23. How have you considered the ethical risks of your solution, and how does your design align with Google's AI Principles?

We consider risks related to bias, misidentification, and responsible use. To mitigate bias, we use diverse datasets spanning species and environments and evaluate performance across contexts. We prioritize transparency through interpretable, trait-based outputs rather than black-box predictions, and incorporate uncertainty estimates to avoid overconfident errors. The system is designed for interactive human-AI collaboration, supporting expert judgment rather than replacing it. We also prioritize data stewardship and responsible use. By expanding equitable access to identification tools and botanical knowledge that are often limited and costly, our approach aligns with Google's AI Principles by promoting scientific benefit, minimizing harm, and ensuring accountability.

24. We expect funded projects to contribute open source outputs for use across the scientific community. Will all outputs funded by Google.org be freely and openly available to the public?

Yes

25. All selected organizations will be invited to join the Google.org Accelerator. How might you leverage Google's technical support and expertise to accelerate project outcomes?

We would leverage the Google.org (<http://Google.org>) Accelerator to transition our prototype into a scalable, robust, and reproducible system. We would benefit from hands-on engineering support to optimize multimodal pipelines, including vision-language trait extraction, retrieval, and re-ranking in Vertex AI, and embedding strategies for fine-grained classification (e.g., Gemini embeddings). We would also benefit from guidance on system architecture, efficient API orchestration, and scaling training and inference on Google Cloud. This support would accelerate the development and deployment of reliable, user-facing tools and improve performance, cost-efficiency, and real-world usability in scientific and environmental applications.

IV. Feasibility

26. Why is your organization or research group uniquely positioned to lead this project?

Our team is uniquely positioned through a combination of deep domain expertise, high-quality ground-truth data, and interdisciplinary collaboration. We have collected and validated extensive field-based ecological datasets over more than a decade, including challenging early life stages where identification is most difficult. Unlike many AI systems trained on public image datasets, our data are expert-validated and embedded within long-term research programs. We maintain established partnerships with the Smithsonian Tropical Research Institute (Panama), New York Botanical Garden, and Fundación para la Conservación de los Andes Tropicales (Ecuador), providing access to complementary data, expertise, and diverse field sites. Our team brings together Yale faculty and interdisciplinary researchers with expertise in botany, ecology, artificial intelligence, and computer vision, positioning us to develop the SeedLearn system, a multimodal framework that integrates structured biological knowledge with advanced AI methods, and to effectively leverage external technical support to scale and deploy these approaches in real-world applications.

27. How would you describe your organization's current AI Maturity?

AI Adoption: AI is used as part of our offerings, either for internally or externally-facing workflows.

28. Describe the work you have done to demonstrate the technical feasibility of your approach. Be sure to describe the specific metrics or evidence you've used to judge success.

We have developed a proof-of-concept multimodal pipeline (SeedLearn) that integrates image embeddings with trait extraction from structured biological sources. Using a curated dataset of ~10,000 expert-validated images, we established baseline benchmarks showing that out-of-the-box models perform poorly (~1% top-1 accuracy) in complex, fine-grained identification tasks. Fine-tuning and trait-informed approaches show consistent improvements in candidate ranking, accuracy, and interpretability. We evaluate performance using top-1 and top-5 accuracy, rare-class performance, confusion matrices, and agreement between extracted traits and expert annotations. The system has been implemented and tested on real-world datasets, showing significant improvements under data scarcity.

29. Please provide an overview of the potential risks (including technical, policy, administrative, or logistical), key dependencies for adoption, and ongoing maintenance requirements associated with your project. For all identified challenges, describe the mitigation strategies you will implement.

Key risks include variability in image quality, extreme data (image and/or textual) scarcity for some classes, and challenges in extracting accurate morphological traits from images. Integrating visual models with structured biological knowledge also introduces technical complexity. To mitigate these risks, we use multiple images per instance, data augmentation, and targeted data collection to improve robustness and coverage. We incorporate uncertainty estimation to flag low-confidence predictions and guide human-AI interaction. Adoption risks include usability in real-world conditions and alignment with user needs. We address this through iterative testing with domain experts and end users, ensuring outputs are interpretable and practical. Key dependencies include continued access to curated data, computational infrastructure, and collaborative networks. We leverage established research partnerships and cloud-based resources to support scalability. Maintenance requirements include updating models with new data and improving performance over time. Our modular system design enables incremental updates, supports integration of improved models, and ensures long-term adaptability as data and technologies evolve.

30. Identify the 3-5 key team members, current or to be hired, that are essential to project success (e.g. software engineer, data analyst).

Ecological Lead: Provides domain expertise in biology, ecology, and trait-based analyses and oversees scientific direction and data collection.

AI/ML Lead: Designs and optimizes multimodal models, focusing on architecture, fine tuning and performance.

Data Scientist: Curates datasets, ensures data quality, manages metadata and conducts benchmarking and statistical evaluation.

Botanical Expert: Validates traits, curates biological knowledge sources, and ensures scientific accuracy.

Technical Engineer: Builds scalable pipelines, infrastructure and user-facing tools for deployment and real-world use.

31. If applicable, please list the partner organisation(s) that you are working with or would like to work with who will support this project proposal's execution and success. [Indicate up to 5 partners]

31.a. Please enter the name of partner organization.

Smithsonian Tropical Research Institute (STRI), Panama

31.b. Please enter your partner organization's website.

<https://stri.si.edu/> (<https://stri.si.edu/>)

31.c. Please share more about the role your partner organization will play in project/expertise.

STRI provides access to long-term tropical forest plots, expert-validated species data, and field infrastructure in Panama. It contributes botanical expertise, supports data collection and validation, enables testing in highly diverse forest systems, and facilitates integration into research workflows, contributing to scaling through adoption across sites and studies.

31.d. Select the option that best describes the relationship you have established with your partner(s).

Existing partner

31.e. Would you like to add another partner?

Yes

31.a.i. Please enter the name of partner organization.

New York Botanical Garden (NYBG), USA

31.a.ii. Please enter your partner organization's website.

<https://www.nybg.org/> (<https://www.nybg.org/>)

31.a.iii. Please share more about the role your partner organization will play in project/expertise.

NYBG contributes taxonomic expertise, access to botanical collections, and structured knowledge from floras and keys. It supports development and validation of trait databases and ensures scientific accuracy. NYBG also enables integration into educational programs and public-facing contexts, contributing to broader adoption and use of SeedLearn.

31.a.iv. Select the option that best describes the relationship you have established with your partner(s).

Existing partner

31.a.v. Add another?

Yes

31.b.i. Please enter the name of partner organization.

Fundación para la Conservación de los Andes Tropicales (FCAT), Ecuador

31.b.ii. Please enter your partner organization's website.

<https://fcatecuador.org/> (<https://fcatecuador.org/>)

31.b.iii. Please share more about the role your partner organization will play in project/expertise.

FCAT supports field implementation of SeedLearn in restoration sites across Ecuador. It contributes local expertise, facilitates data collection in diverse ecosystems, and enables real-world testing in conservation practice. Through its network, FCAT connects the project with practitioners, supporting broader application and scaling across restoration contexts.

31.b.iv. Select the option that best describes the relationship you have established with your partner(s).

Existing partner

31.b.v. Add another?

No

V. Scalability

32. How will you scale or build upon your core idea? Detail how success could expand beyond the initial proposal.

- Geographic transfer (e.g. moving from a local pilot to a regional or global scale)
- Exponential user growth (e.g. broad consumer adoption)
- Technical & performance maturity (e.g. optimizing for enterprise-scale throughput)
- Ecosystem & integration (e.g. becoming a foundational tool for other developers)

33. Address how the team's structure and capabilities would need to evolve to support this growth at scale. If you're expecting a partner organization to scale your idea, please explain.

The system is designed to scale across species, sites, regions, and application domains. To support this growth, the team will expand capacity in AI/ML engineering, data infrastructure, and software development, with increased focus on system deployment, maintenance, and user-facing tools. The team structure will evolve from research-focused development to include engineering and application-focused roles that support practical use. As the system matures, the team will increasingly focus on usability and integration into existing platforms and workflows. We expect partner organizations to play a key role in scaling through co-development, feedback, and adoption, applying the system across diverse field contexts and embedding it within existing practices such as research, teaching, and restoration planning.

34. Google.org generally makes one-time contributions. How would you financially and technically sustain this work beyond Google.org's support?

34.a. Financial sustainability

We will sustain this work through a combination of research grants, institutional support, and partnerships with conservation and research organizations. We will explore potential service-based applications (e.g., monitoring tools) and collaborations with NGOs and government agencies to support continued deployment, maintenance, and long-term impact.

34.b. Technical sustainability

The system will use a modular architecture and widely adopted tools to support ongoing development and adaptability. Models and datasets will be maintained through institutional infrastructure and shared repositories. Versioned pipelines and partnerships with research and applied organizations will enable continued data integration, validation, and iterative model improvement.

35. What key learnings will your project generate, and how will you share them with other organizations to help advance the field?

This project will generate insights on integrating domain knowledge with AI in image-limited settings, improving fine-grained identification, and combining visual and trait-based reasoning. We will evaluate how multimodal approaches improve performance, interpretability, and usability in real-world contexts. We will share methods, models, and best practices through publications, conference presentations, and, where appropriate, accessible repositories and pipelines. We will also document design principles for human–AI collaboration and multimodal systems, supporting broader adoption across scientific, environmental, and applied domains.

36. If applicable, please provide information related to your organization's public presence and project visibility.

This project builds on prior work supported by a Yale AI Seed Grant (<https://ai.yale.edu/seed-grant-awards-2025> (<https://ai.yale.edu/seed-grant-awards-2025>)), which funded development of our SeedLearn pipeline for AI-based seedling identification. Related work has been presented at the Yale AI Symposium (<https://ai.yale.edu/symposium-2025> (<https://ai.yale.edu/symposium-2025>)) and in academic venues. The team combines expertise in ecology, botany, and AI, with ongoing collaborations across research institutions (see <https://nohemihuanca.github.io/seedlearn/> (<https://nohemihuanca.github.io/seedlearn/>)). We have an existing proof-of-concept integrating image data with structured biological knowledge, demonstrating feasibility and forming the foundation for this project.

VI. Project Budget and Timeline

For the questions below, we are asking for your early assessment based on the information you have today. The budget table and activities are not expected to be final, and selected finalists will ultimately undergo an iterative proposal development

process with the Google.org (<http://Google.org>) team.

37. Funding Request: The GIC expects to award funding between \$500K and \$3M USD. Please request a funding amount [USD] that most closely matches your project's needs.

\$1,500,000

Budget Breakdown: For each of your 3 to 5 high-level categories, please answer the following three prompts. Please focus on the most significant costs that directly impact the execution and success of your project, and are readily identifiable as being linked to a specific activity. Common categories may include:

- **Personnel & staffing:** project team salaries, especially technical roles like developers and data scientists; level of specificity expected is, for example, "Data Scientist, 1, 24 months, \$X total cost"
- **Technology development:** costs for software, hardware, data acquisition, and development of AI models
- **Equipment & Infrastructure:** costs for compute and cloud services
- **Project amplification & other business costs:** activities that foster collaboration; any subawards to partners expected to play a substantial role in the project, travel and conferences, scientific journal fees, institutional overhead cost, staff training, etc
- **Measurement & evaluation:** resources for measuring the project's success and impact
- **Indirect Costs:** rent, utilities, etc (Typically limited to 10-12% of total budget).

38. Budget Category 1

38.a. Budget Category Name

Personnel & Staffing

38.b. Budget Allocation (USD)

\$934,299.64

38.c. Description & Subcategory Details

Supports a focused, interdisciplinary team essential for developing and deploying the SeedLearn system over a 3-year period. Budget includes a full-time AI postdoctoral researcher (100% effort) leading model development; partial support for the research scientist (ecology lead, project lead, 50% effort) ensuring biological grounding and data integration; a data scientist (30% effort) supporting data curation and evaluation; and a PhD student contributing to experimentation and analysis. Targeted effort is allocated to three Yale faculty Co-PIs, who provide critical scientific and technical leadership with complementary expertise in AI, computer vision, and ecology. This structure ensures efficient, high-impact use of resources.

39. Budget Category 2**39.a. Budget Category Name**

Technology Development (AI Models & APIs)

39.b. Budget Allocation (USD)

\$154,545

39.c. Description & Subcategory Details

Supports the use of large language models (LLMs) and embedding APIs for multimodal inference and reasoning within the SeedLearn system. These costs cover model inference, embedding generation, and retrieval-augmented workflows, enabling integration of structured biological knowledge with image-based models.

These resources are essential for trait extraction, candidate ranking, and scalable species identification in data-limited environments where traditional approaches are insufficient.

40. Budget Category 3

40.a. Budget Category Name

Compute & Infrastructure

40.b. Budget Allocation (USD)

\$185,454

40.c. Description & Subcategory Details

Covers cloud-based GPU compute and storage required for training, scaling, and deploying multimodal models on large, heterogeneous datasets. These resources support large-scale data processing, model training, cross-site generalization experiments, and efficient data pipelines. This infrastructure enables integration of multimodal data at scale and ensures robust performance, scalability, and reproducibility across diverse environmental conditions.

41. Budget Category 4

41.a. Budget Category Name

Data Expansion & Field Validation

41.b. Budget Allocation (USD)

\$61,818

41.c. Description & Subcategory Details

Supports targeted data collection and validation across diverse environments to expand dataset coverage and improve representation of rare and under-sampled species. This effort builds on existing long-term research sites with expert-verified species identification, providing a high-quality ground-truth dataset that is rare in highly diverse systems. Costs include additional image acquisition, metadata collection, and collaboration with local partners to extend coverage across conditions. These efforts enable robust evaluation across sites, improving model generalization, accuracy, and performance under real-world ecological variability.

42. Budget Category 5

42.a. Budget Category Name

Travel, Dissemination & Project Support (including indirect costs)

42.b. Budget Allocation (USD)

\$223,448.35

42.c. Description & Subcategory Details

Covers travel for conferences, collaboration meetings, and engagement with scientific and applied partners to support dissemination, adoption, and real-world deployment. Includes costs for presenting results, stakeholder engagement, and integration into research and applied workflows. This category also supports open science outputs, including publications, datasets, and tools, as well as institutional overhead (15%). These resources ensure effective project coordination, visibility, and long-term impact across scientific, environmental, and applied domains.

Project Timeline Overview

Please provide an overview of your project's key activities and expected outcomes, including 3-5 key milestones, by completing the questions below. Funding can be used over the course of 36 months, including the Google.org (<http://Google.org>) Accelerator period. Clearly indicate the timeframe for each row.

43. Project Milestone 1

43.a. Timeframe

Foundation and Data Integration (Months 1–6)

43.b. Activities

Build on the existing SeedLearn pipeline by refining and consolidating curated image datasets and associated metadata. Improve taxonomic standardization and cross-dataset species alignment, and strengthen links between visual data and structured biological knowledge sources. Develop and refine the RAG system to improve the reliability of trait retrieval and reasoning across diverse sources. Refine preprocessing pipelines, develop traits schema, and integrate multi-view data, and establish baseline benchmarks using the multimodal framework for identification accuracy, trait extraction, and interpretability

43.c. Outcomes/Key Milestones

This phase will produce an integrated multimodal dataset linking images with structured biological knowledge, along with an improved RAG system for reliable trait extraction and reasoning. We will establish baseline benchmarks for species identification and trait inference, including quantitative metrics (e.g., top-1/top-5 accuracy and agreement with expert annotations). We will also develop a clear evaluation framework to assess accuracy, interpretability, and generalization across datasets and environmental conditions, providing a measurable foundation for subsequent model refinement and scaling.

44. Project Milestone 2

44.a. Timeframe

Model Refinement and Trait-Based Reasoning (Months 7–12)

44.b. Activities

Refine the existing multimodal framework by improving integration between image embeddings and language-based trait extraction. Enhance detection of key morphological traits from field images and strengthen the RAG system for more accurate linkage between visual features and botanical descriptions. Conduct systematic comparisons between image-only and trait-informed models relative to baseline performance, and quantify improvements in identification accuracy and interpretability. Improve model calibration and incorporate uncertainty estimation to better handle ambiguous or low-quality inputs, enabling more reliable and transparent decision-making in real-world conditions.

44.c. Outcomes/Key Milestones

This phase will deliver measurable improvements in model performance relative to baseline benchmarks, including increases in top-1 and top-5 identification accuracy in image-limited scenarios. We will demonstrate gains in interpretability through validated trait-based reasoning, measured by agreement with expert-annotated traits. Comparative evaluations will confirm improvements over image-only approaches. In addition, we will produce a more robust and calibrated multimodal system, with reduced prediction uncertainty and improved reliability across varying image quality and environmental conditions.

45. Project Milestone 3

45.a. Timeframe

Scaling and Cross-Site Generalization (Months 13–20)

45.b. Activities

Expand datasets to include additional species, environments, and imaging conditions across sites, increasing taxonomic and environmental diversity. Integrate new multi-view field images and associated metadata, and standardize data pipelines for cross-site compatibility. Optimize training and inference pipelines for scalability using cloud-based infrastructure (e.g., distributed training, efficient data loading, and batch processing). Evaluate model performance across diverse contexts, including cross-site and cross-species generalization. Improve robustness to variations in image quality, lighting, and environmental conditions by leveraging targeted data augmentation and model regularization, ensuring consistent performance across real-world, heterogeneous datasets.

45.c. Outcomes/Key Milestones

This phase will demonstrate generalization across datasets and environmental conditions, quantified through consistent performance across sites and species not included in training. We will measure improvements in identification accuracy and trait extraction in complex, data-scarce scenarios, and validate robustness under varying image quality and field conditions. Scalable training and inference pipelines will be implemented and benchmarked for efficiency and throughput. Results will provide clear evidence that multimodal, trait-based approaches outperform image-only models in fine-grained identification tasks across diverse ecological contexts.

46. Project Milestone 4

46.a. Timeframe

Deployment and User-Focused System Development (Months 21–28)

46.b. Activities

Develop and refine user-facing tools (e.g., web interface or application), building on the existing prototype, enabling multi-image input and interactive querying. Integrate interpretable outputs, including trait-level explanations, confidence scores, and alternative candidate species. Conduct structured pilot testing with researchers, students, and practitioners across multiple field contexts, collecting quantitative and qualitative feedback on usability, accuracy, and decision support. Implement iterative improvements to interface design, model performance, and response clarity based on user feedback. Evaluate system integration within real-world workflows, including field identification and data collection tasks, to ensure usability under practical conditions.

46.c. Outcomes/Key Milestones

This phase will deliver a fully functional, user-tested system with demonstrated usability across real-world workflows. We will quantify user performance improvements, including reductions in identification time and increased accuracy relative to baseline workflows. User studies will measure trust, usability, and effectiveness of trait-based explanations in supporting decision-making. The system will provide calibrated confidence estimates and interpretable outputs validated

through user feedback. Results will demonstrate effective human–AI interaction and produce a refined, deployable system ready for broader testing and adoption in research and applied settings.

47. Project Milestone 5

47.a. Timeframe

Validation, Dissemination, and Impact (Months 29–36)

47.b. Activities

Conduct comprehensive validation using independent datasets spanning new species, sites, and environmental conditions to assess generalization beyond training data, including evaluation on unseen species and cross-site transfers. Evaluate system performance, interpretability, and robustness in real-world applications, including biodiversity monitoring and restoration workflows. Perform systematic benchmarking against existing image-only and multimodal approaches using standardized evaluation protocols. Finalize model improvements, system optimization, and documentation, including reproducible pipelines, open-source-ready code, and user guidelines for public release. Disseminate results through peer-reviewed publications, conferences, and open-access platforms. Release datasets, models, and tools, and deploy accessible interfaces (e.g., web tools or APIs), while engaging partner organizations to support adoption, integration into existing workflows, and long-term sustainability.

47.c. Outcomes/Key Milestones

This phase will deliver a validated, scalable multimodal AI system with demonstrated performance across independent datasets and real-world applications. We will report quantitative metrics for identification accuracy, trait extraction reliability, and generalization across sites and species. Benchmarking will provide clear evidence of improvements over image-only and alternative approaches. Outputs will include peer-reviewed publications, publicly available datasets, reproducible pipelines, and accessible tools. Adoption by partner organizations and engagement metrics (e.g., users, deployments) will demonstrate real-world impact, supporting biodiversity monitoring, ecological research, and applied environmental decision-making across diverse contexts.

VII. Ethics and Compliance

Ethics and Compliance: Please answer the following Ethics & Compliance questions carefully and provide additional information if any of the questions below apply to you, your organization, and any partner organizations. If you indicate "Yes" to any of the questions below, or if you're unsure about your responses, please provide as much detail as possible in question 53.

48. To your knowledge, does your organization have any ongoing commercial contracts or negotiations with Google that are directly related to the project detailed in this application?

No

49. To your knowledge, are any current government officials, employees or civil servants employed by your organization or are they members of your organization's board or officer group? If you are employed by a public university, please select yes.

No

50. To your knowledge, are any current government or intergovernmental entities, officials, employees, or civil servants (collectively "Government Persons") involved in your project?

Yes

51. If selected to receive funding, will your organization engage with or provide funding to any local, state, federal (including military) law enforcement entities or officials on the project?

No

52. Does your organization have any dealings or programs in Crimea, Cuba, Iran, North Korea, Russia, the so-called Donetsk People's Republic (DNR) and Luhansk People's Republic (LNR), or Syria?

No

53. If you answered "yes" to any of the above questions, please explain here. If not, enter N/A.

The project includes collaboration with researchers affiliated with a public-private institution, specifically the Smithsonian Tropical Research Institute in Panama, which is a branch of the Smithsonian Institution. However, no project funds will be used to pay salaries, stipends, travel, or other costs of federal government employees. Smithsonian federal employee involvement is limited to scientific collaboration and advisory input. Funds to cover research technicians' salaries for data collection in Panama will run through the Smithsonian Tropical Research Institute, but these technicians are employed locally in Panama and are not government employees.

Certification & Completion

By submitting my Application, I certify on behalf of my organization that: (1) my organization and the points of contact listed in my Application have consented and agreed to be contacted by Google for purposes of evaluating and potentially funding the project described in my Application; (2) my organization agrees to the [Terms](https://services.google.com/fh/files/blogs/gic_aisci_terms.pdf) (https://services.google.com/fh/files/blogs/gic_aisci_terms.pdf); (3) all information contained in this Application is true and accurate; (4) my organization meets the eligibility requirements specified in the [Terms](https://services.google.com/fh/files/blogs/gic_aisci_terms.pdf) (https://services.google.com/fh/files/blogs/gic_aisci_terms.pdf); (5) none of the information in this Application is confidential or proprietary; and (6) I am an authorized representative of my organization who has the authority and permission from my organization to submit an Application on behalf of the organization and has the authority to legally bind the organization or have permission from an authorized representative to submit an Application on behalf of the organization, – including, without limitation, to the [Terms](https://services.google.com/fh/files/blogs/gic_aisci_terms.pdf) (https://services.google.com/fh/files/blogs/gic_aisci_terms.pdf).

Note: The information you submit will be used in accordance with Google's Privacy Policy.

Yes, I certify

You are about to submit your application.

Correcting mistakes or making edits after your application has been submitted is not allowed. For any technical questions about the platform, please visit [THIS WEBSITE](https://submittable.help/en/collections/185534-help-for-applicants) (<https://submittable.help/en/collections/185534-help-for-applicants>).

To complete and submit your application click 'Submit'. Your email service provider will sometimes try to predict incoming spam mail on your behalf and filter it into your spam folder before you ever see it. If you're not receiving expected emails from notifications@email.submittable.com (<mailto:notifications@email.submittable.com>), you will want to safelist the address to help ensure that these emails reach you.

Safelisting an email address means explicitly telling your email service provider that emails from notifications@email.submittable.com (<mailto:notifications@email.submittable.com>) are not spam or junk mail. [How can I safelist notifications from Submittable?](https://submittable.help/en/articles/3221476-how-can-i-safelist-notification-emails-from-submittable) (<https://submittable.help/en/articles/3221476-how-can-i-safelist-notification-emails-from-submittable>).